

# Wen-Chieh Chao, MD MS

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Ultrasound-responsive biomaterials • 3D bioprinting • Engineered cardiac tissues

## RESEARCH EXPERIENCE

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### Graduate Researcher

Oct 2024 – Present

*The Skylar-Scott Lab, Stanford - Biomaterials, Liposomes*

*Palo Alto, CA*

- Enabled controllable, ultrasound-triggered fibrin crosslinking by engineering a thrombin-loaded, ultrasound-responsive liposome release platform.
- Achieved robust and tunable thrombin release by optimizing membrane mechanics across >100 liposome fabrication conditions.
- Made evaluation of effects on lab-engineered growth factor possible by conducting iPSC culture experiments.

### Research Assistant

Dec 2023 - Jun 2024

*Wu Lab, IMB Academia Sinica - Neuroanalysis, Neuro-behavior*

*Taipei, Taiwan*

- Characterized DBS-induced behavioral and neural changes in Parkinson's mouse models by conducting controlled behavioral assays on >20 PD mice.
- Increased reproducibility and throughput of dendritic spine quantification by optimizing DeepD3-based analysis workflows.
- Extracted nonlinear structure in neural datasets by using PCA/UMAP-based dimensionality-reduction analyses.

### Research Assistant

Apr 2021 - Jun 2021

*Brain and Mind Lab, National Taiwan University - FMRI, VR design*

*Taipei, Taiwan*

- Identified age-related differences in brain activation by performing full FMRI preprocessing and 2nd-level SPM analysis in MATLAB.
- Enabled controlled testing of spatial navigation behavior by building non-Euclidean virtual environments in Unity.
- Reduced experimental analysis time by 80% by integrating open-source gaze-analytics pipelines into the workflow.

### Rotation Internship

Oct 2022 - Nov 2022

*The Tytgat Institute, Amsterdam UMC - GPCR, Transfection*

*Amsterdam, Netherlands*

- Screened bile-salt-responsive GPCR candidates by applying the Presto-Tango reporter platform in HTLA cells.
- Boosted transfection success rate from 50% to 90% by optimizing reagent stoichiometry and workflow conditions.
- Supported project planning by synthesizing bile-salt signaling literature & presenting updates to research team.

## PUBLICATIONS & PRESENTATIONS

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Manuscript in preparation.

## TECHNICAL SKILLS

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**Wet Lab & Biomaterials:** Liposome fabrication, protein encapsulation, iPSC culture, hydrogel/fibrin crosslinking

**Computational & Modeling:** MATLAB, Python, FMRI/SPM, neural signal analysis, PCA/UMAP

**Instrumentation & Imaging:** Ultrasound transducers, confocal/fluorescence microscopy, FIJI, DeepD3

## EDUCATION

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### Stanford University

2024 - 2026

*Masters of Science - Biomedical Engineering*

*Palo Alto, CA*

- Research focus: ultrasound-responsive biomaterials and 3D bioprinting.
- GPA: 3.91

### National Taiwan University

2017 – 2023

*Doctor of Medicine - MD, Medicine*

*Taipei, Taiwan*

- Clinically trained with full two-year hospital rotations.
- GPA: 3.98